

Joshua Kennedy

Platform Software Engineer
Resides in Scotland



Experience

Cloud Infrastructure Engineer

NatWest - October 2023 - Present

- > Engineered and supported strategic on-premise Container as a service (CaaS) capabilities used by internal engineering teams.
- > Applied platform engineering principles to improve shared OpenShift and Kubernetes-based products, with a focus on reliability, automation, availability and developer experience.
- > Developed platform engineering capabilities for OpenShift through policy-driven automation and GitLab CI/CD, enabling GitOps orchestration, HashiCorp Vault integration and customer guardrails.
- > Worked with principal engineers, architects, platform teams and stakeholders to understand platform requirements and deliver pragmatic technical outcomes within a regulated financial services environment.
- > Continuously evaluated and prototyped improvements to platform tooling, automation and operational workflows.

Cloud Automation Engineer

NatWest - September 2021 - October 2023

- > Joined a small R&D team focused on expanding the bank's on-premise cloud computing capabilities.
- > Built experience across OpenShift, Kubernetes, container orchestration, Linux and cloud platform automation.
- > Delivered proof-of-concept work for certificate automation using cert-manager and centralised secret management using HashiCorp Vault.
- > Collaborated across teams to turn emerging platform requirements into working prototypes and executable patterns.
- > Supported automation and CI/CD-aligned delivery practices to reduce manual effort and increase consistency across platform operations.

Middleware Systems Engineer

NatWest - February 2019 - September 2021

- > Built automation to streamline common support tasks for deployment orchestration tooling within a feature team.
- > Automated agent upgrade workflows, enabling over 1,600 agents to be upgraded across approximately 50 teams within six months.
- > Supported users and administrators of deployment orchestration platforms, producing documentation and running workshops to upskill a wider support team.
- > Improved platform support processes by combining scripting, documentation, incident analysis and close collaboration with consuming teams.

Middleware Technical Analyst

RBS - January 2018 - February 2019

- > Delivered technical support for Java-based enterprise middleware, resolving over 150 individual incidents.
- > Developed infrastructure support scripts and helped stand up deployment orchestration tooling.
- > Built practical experience with REST API integration, high-availability clustering and production support in a regulated banking environment.

WebSphere Technical Apprentice

RBS - January 2016 - January 2018

- > Developed KornShell automation to replace a manual licensing process across approximately 5,000 servers.
- > Audited and improved certificate renewal tracking workflows, building skills in Visual Basic, SQL and enterprise middleware administration.
- > Completed Modern and Graduate Apprenticeship pathways while working in production financial services technology teams.

Education and Certification

Certificate of Higher Education in Engineering

April 2023 - October 2025

Open University. Completed four engineering modules with Distinction, strengthening mathematical, analytical and systems engineering foundations.

Red Hat OpenShift Administration II: Configuring a Production Cluster (DO280)

2024

Production OpenShift cluster administration, configuration and operational practices.

Red Hat Certified Specialist in Containers and Kubernetes

July 2022

Proficiency in Podman, Kubernetes and OpenShift.

Graduate Apprenticeship Scotland in IT & Telecommunications Professionals

January 2017 - November 2017

SCQF Level 8-11 Diploma for IT & Telecommunications Professionals, QA Apprenticeship.

Modern Apprenticeship Scotland in IT & Telecommunications Professionals

January 2016 - November 2016

SCQF Level 6 Diploma for IT & Telecommunications Professionals, QA Apprenticeship.

HND Computer Games Development

August 2013 - August 2015

Edinburgh College. Covered object-oriented programming, web development, abstract data structures, project work and user interface design.

Personal Projects

tapo-onboarding

Shell / Reverse Engineering / IoT

Open-source tooling for onboarding modern TP-Link Tapo cameras without using the vendor cloud, including support for newer v3 encrypted provisioning.

Stateful

JavaScript / Pebble / REST APIs

Pebble smartwatch application providing a tile-based interface for controlling RESTful APIs with minimal user interaction.

restate-go

Go / Home Automation / APIs

Local-first home automation platform written in Go, designed to provide cloudless automation without dependence on external services.

Snowdon II WiFi

C++ / Raspberry Pi Pico / REST APIs / Infrared

Raspberry Pi Pico based modchip adding Wi-Fi and a RESTful API to a Majority Snowdon II soundbar by driving its infrared interface and inferring state from status LED.

About me

Platform and software engineer with almost ten years' experience delivering infrastructure, middleware and cloud platform capabilities in financial services. I specialise in using software engineering, automation and platform engineering principles to reduce operational complexity, improve delivery speed and make shared platforms easier for other engineers to consume. My background spans OpenShift, Kubernetes, Linux, middleware, CI/CD, REST APIs, scripting and developer tooling, with practical software development experience across Go, Shell, JavaScript and C++. Alongside my professional work, I build and maintain open-source projects involving cloudless device onboarding, local-first home automation, embedded systems and API-driven hardware control.

Skills

Platform Engineering

OpenShift / Kubernetes

Linux

Automation

Shell / Bash / KornShell

Go

Python

JavaScript

REST APIs

CI/CD

Infrastructure as Code

Security and Policy Controls

Stakeholder Communication

Embedded C / C++

Contact



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